Surface EMG-Based Hand Gesture Recognition

Farzan Aghaei · Sanaz Mazrae

Department of Electrical Engineering, Sharif University of Technology | Sensing and Measurement

Abstract

This project investigates recognition of hand and wrist movements from multichannel surface electromyography (sEMG). A recording from Subject 14, Exercise 1 of the NinaPro DB2 dataset is analyzed using twelve EMG channels sampled at 2,000 Hz. Signals are band-pass filtered between 20 and 450 Hz and partitioned into 200 ms windows with a 100 ms stride, producing ten handcrafted descriptors per channel. Four supervised classifiers—support vector machine (SVM), linear discriminant analysis (LDA), K-nearest neighbors (KNN), and random forest—are evaluated on movement repetitions held out from training. Across 17 gesture classes, random forest achieved the best performance: 74.26% accuracy, 75.09% balanced accuracy, and 75.17% macro-F1. The experiment demonstrates within-participant gesture discrimination while maintaining independence between training and evaluation repetitions.

1. Project Objective and Scope

The objective is to classify hand and wrist movements from short, multichannel EMG recordings. The analysis comprises signal filtering, segmentation, handcrafted feature extraction, supervised classification, and evaluation on held-out movement repetitions.

The task contains 17 non-rest movements from one participant. Each analysis window receives a single gesture label; evaluation concerns unseen repetitions within the same participant, rather than performance on unseen people.

2. Recording and Experimental Data

The input file S14_E1_A1.mat contains 1,804,139 EMG samples per channel (approximately 902.1 s). Corrected sample-level gesture and repetition annotations are provided by restimulus and rerepetition. The analyzed movements have six repetition identifiers; rest and unassigned samples are excluded.

Parameter	Value
Dataset	NinaPro DB2
Recording	Subject 14 · Exercise 1
EMG channels	12
Sampling frequency	2,000 Hz
Raw samples	1,804,139
Analyzed labels	17 gesture classes (1–17)
Sample labels	restimulus / rerepetition
Excluded samples	Rest class 0; unassigned repetitions

Table 1. Dataset specification and label definition.

3. Signal Preprocessing

All channels undergo the same fourth-order Butterworth band-pass filter with cutoff frequencies of 20 and 450 Hz. A forward–backward, second-order-section implementation is used to avoid net phase shift in the processed

waveform. No additional notch filter is included in this experiment. The filter targets the sEMG activity band while attenuating low-frequency baseline variation and higher-frequency components outside the selected passband.

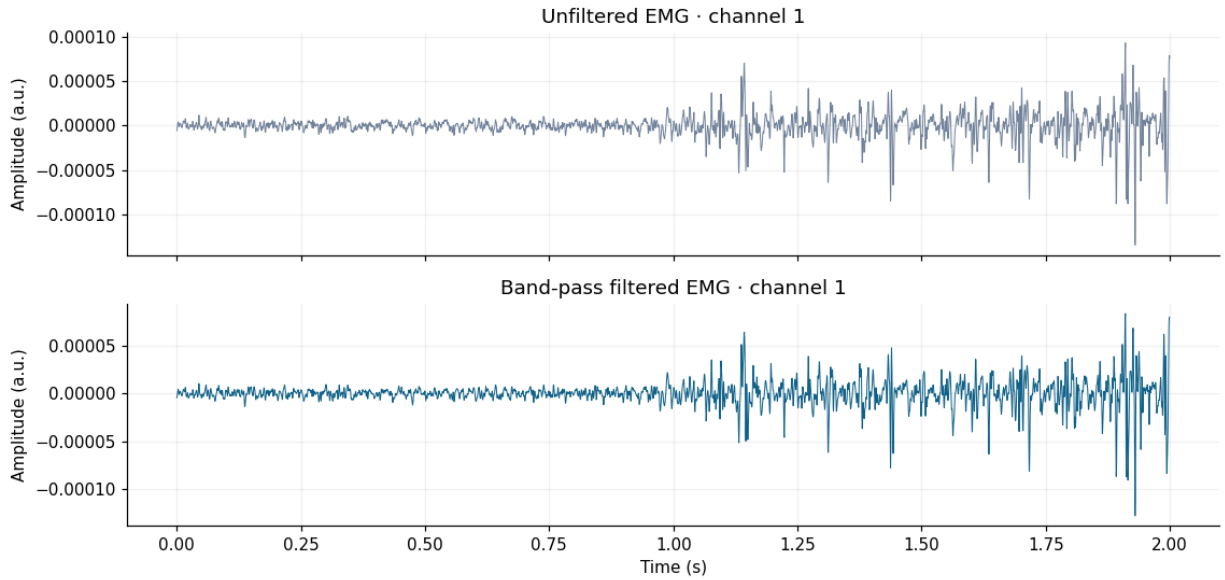


Figure 1. Representative raw and band-pass filtered waveform from EMG channel 1 (same 2 s interval).

Figure 1 demonstrates the difference between the acquired signal and the conditioned waveform used for feature extraction. To evaluate the filtering in the frequency domain, Welch power spectral density estimates are computed for channel 1 of the complete recording.

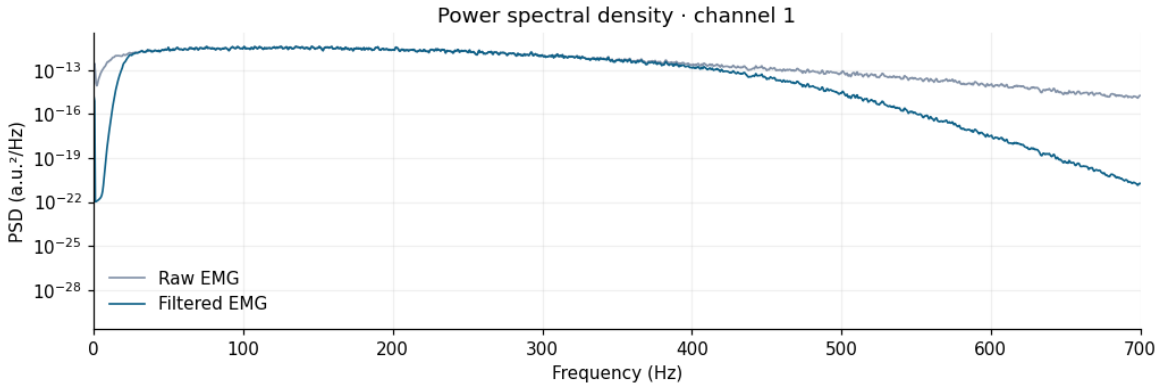


Figure 2. Power spectral density of the first EMG channel before and after band-pass filtering.

4. Windowing and Feature Representation

4.1 Movement-consistent segmentation

The filtered EMG is divided into 200 ms windows (400 samples) with a 100 ms stride (200 samples), corresponding to 50% overlap within a movement segment. Boundaries are introduced at every change in either the corrected movement label or repetition identifier. Windows are constructed only within these contiguous segments. Consequently, no feature vector combines observations from different movements or repetitions, and no window assigned to one held-out repetition is used for model fitting.

The segmentation yields 4,698 labeled windows. Each window contains 12 channels and is represented by a fixed-length 120-component vector: ten descriptors for each channel. Feature values are calculated directly from the filtered waveform without dimensionality reduction or learned feature extraction.

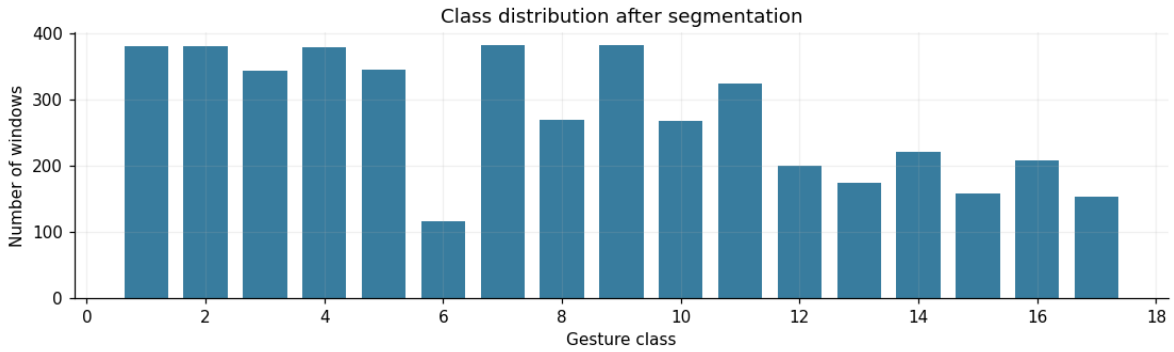


Figure 3. Number of retained analysis windows per gesture class.

4.2 Extracted signal descriptors

The features summarize signal amplitude, temporal variation, event counts, and correlation between neighboring EMG channels. Let x[n] denote a channel waveform in a window containing N = 400 samples; the operators diff(·), std(·), and var(·) follow the NumPy definitions used in the notebook.

Feature	Calculation / interpretation
MAV	Mean absolute value: $(1/N)\,\Sigma\,\lvert x[n]\rvert$.
STD	Standard deviation of the sampled window.
VAR	Population variance of window samples.
WL	Waveform length: $\Sigma\,\lvert x[n] - x[n-1]\rvert$.
ZC	Count of sign changes across adjacent samples, subject to an amplitude-change threshold.
RMS	Root mean square: $\sqrt{[(1/N)\,\Sigma\,x[n]^2]}$.
NP	Number of peaks in $\lvert x[n]\rvert$ above the specified prominence threshold.
SSC	Count of changes in successive difference signs, with a difference threshold.
CORR	Pearson correlation between the current and preceding EMG channel.
IAV	Integrated absolute value: $\Sigma\,\lvert x[n]\rvert$.

Table 2. Ten time-domain descriptors computed for every EMG channel.

Thresholds reduce spurious event counts: zero-crossing and slope-sign thresholds use $\max(10^{-10}, 0.01 \times$ channel STD). Peak detection operates on the rectified waveform with prominence $\max(10^{-10}, 0.25 \times$ channel STD). For CORR, channel 1 uses a fixed zero placeholder; later channels use the immediately preceding channel, with zero returned if the correlation cannot be determined reliably. MAV/IAV and STD/VAR are related but retained as distinct components of the specified feature representation.

5. Model Training and Experimental Protocol

5.1 Training and test partitions

Repetitions 2 and 5 are held out for testing; all other repetitions form the training set. Partitioning is performed by repetition ID. No repetition contributes windows to both partitions, and all classifiers receive the same features and test samples.

Partition	Repetitions	Windows
Training	1, 3, 4, 6	3,183
Testing	2, 5	1,515
Total	1–6	4,698

Table 3. Repetition-disjoint training and test partition.

5.2 Classifier specifications

SVM, LDA, and KNN use z-score scaling fitted exclusively to training windows; random forest uses unscaled features. The held-out test set is not used for tuning.

Model	Configuration	Scaling
SVM	RBF kernel; C = 1; gamma = scale	StandardScaler
LDA	LinearDiscriminantAnalysis; SVD solver	StandardScaler
KNN	k = 5; Euclidean distance	StandardScaler
Random Forest	100 trees; random_state = 42	None

Table 4. Classifier hyperparameters and training-time preprocessing.

5.3 Performance criteria

Accuracy is the percentage of correctly classified windows. Balanced accuracy averages recall across the 17 gestures; macro-F1 equally averages their class-specific F1 scores. Confusion matrices tabulate true labels by row and predicted labels by column.

6. Experimental Results

6.1 Overall model comparison

Table 5 summarizes evaluation on 1,515 held-out windows. Random forest achieves the highest scores across all metrics; SVM ranks second, followed by LDA and KNN.

Classifier	Accuracy (%)	Balanced accuracy (%)	Macro-F1 (%)
Random Forest	74.26	75.09	75.17
SVM (RBF)	67.72	70.31	70.29
LDA	62.77	65.69	66.04
KNN (k=5)	61.45	64.65	65.49

Table 5. Performance on the held-out repetition test set.

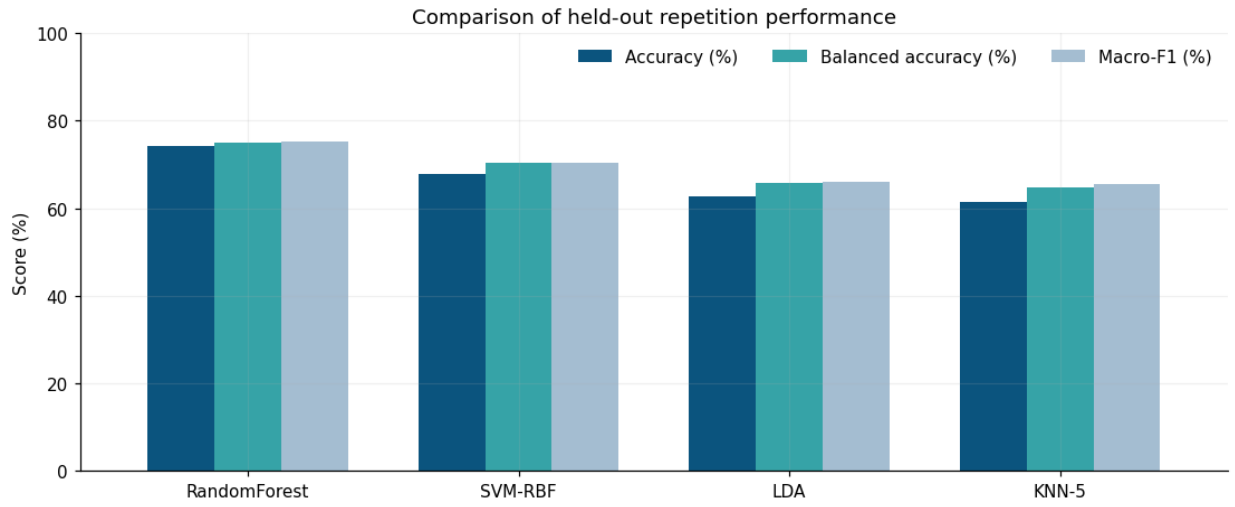


Figure 4. Comparison of accuracy, balanced accuracy, and macro-F1 over the same test set.

The random forest improves overall accuracy by 6.54 percentage points relative to the SVM (74.26% versus 67.72%). Its macro-F1 score of 75.17% indicates that the advantage is not restricted to the aggregate number of

correctly predicted windows. The ranking is based on this recording and feature configuration, without claiming universal superiority across other subjects or datasets.

6.2 Confusion-matrix analysis

The confusion matrices in Figure 5 display the distribution of errors across the seventeen gesture classes. Diagonal entries indicate correctly classified windows and off-diagonal entries indicate confusion between gestures. Stronger diagonal structure is visible for the random forest, while several classes retain overlapping patterns under every model.

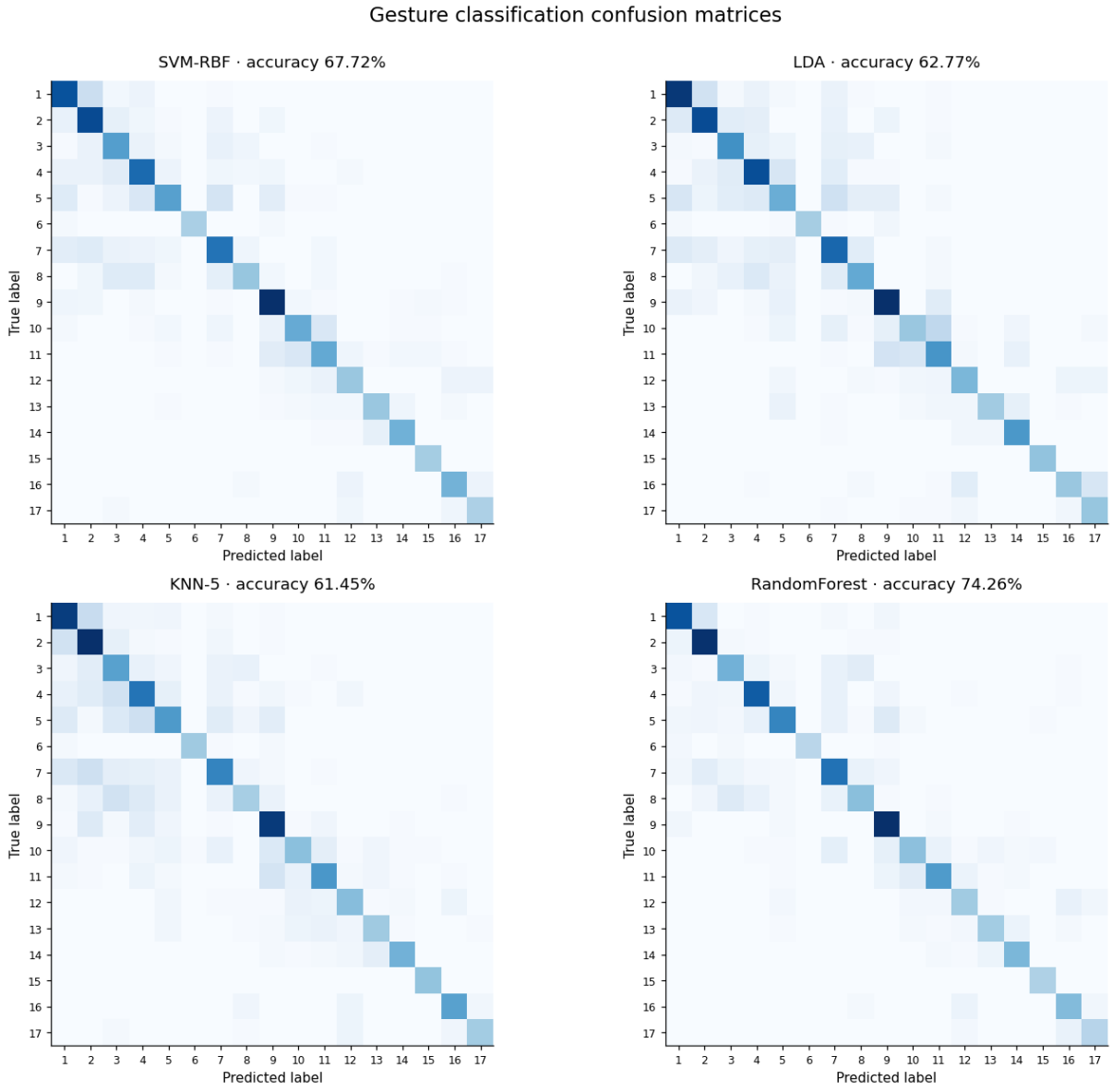


Figure 5. Confusion matrices for the SVM, LDA, KNN, and random-forest models; identical test windows and class ordering.

6.3 Class-specific behavior

The best-performing classifier is examined at the gesture level using per-class F1 scores. The resulting variability shows that classification difficulty differs between movement labels even when the same preprocessing, features, and evaluation split are used.

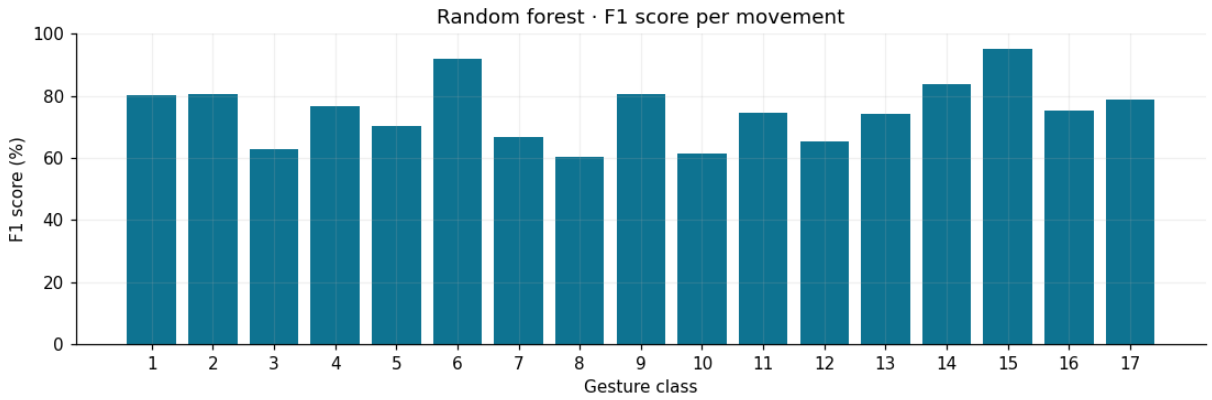


Figure 6. Per-class F1 scores of the random-forest model on held-out repetitions.

Table 6 provides the numerical class-wise evaluation. The number of test windows differs by gesture, which is important when interpreting a high or low F1 score for an individual movement.

Gesture	Precision (%)	Recall (%)	F1 (%)	Test windows
1	80.31	80.31	80.31	127
2	71.78	91.41	80.41	128
3	64.04	61.96	62.98	92
4	76.56	76.56	76.56	128
5	81.44	61.72	70.22	128
6	100.00	85.00	91.89	40
7	64.71	68.75	66.67	128
8	66.67	54.95	60.24	91
9	71.08	92.91	80.55	127
10	71.01	53.85	61.25	91
11	80.23	69.70	74.59	99
12	65.15	65.15	65.15	66
13	76.36	72.41	74.34	58
14	81.82	85.71	83.72	63
15	90.48	100.00	95.00	38
16	71.23	80.00	75.36	65
17	81.40	76.09	78.65	46

Table 6. Gesture-level precision, recall, F1 score, and test support for the random forest.

The highest F1 values occur for classes 15 (95.00%, 38 test windows) and 6 (91.89%, 40 windows). The lowest are for classes 8 (60.24%, 91 windows) and 10 (61.25%, 91 windows). These results indicate that discrimination difficulty and the amount of evaluation data differ across movements.

The most frequent off-diagonal errors in the random-forest matrix are listed in Table 7. These values are window counts, not percentages; they identify the true movement followed by the incorrectly predicted movement.

True class	Predicted class	Misclassified windows
1	2	18
8	3	15
5	9	15
11	10	13
3	8	13

Table 7. Five highest off-diagonal error counts for the random forest.

7. Discussion and Conclusion

The results establish that a compact, interpretable feature representation can separate multiple hand and wrist movements from a single participant's sEMG. Incorporating amplitude statistics, waveform variation, peak and crossing counts, and between-channel correlation provides a common representation for four different learning algorithms. With an identical train/test protocol, random forest achieves the strongest combination of overall accuracy and class-balanced metrics.

The test design explicitly separates movement repetitions. This choice makes the assessment more representative of within-subject recognition on temporally distinct performances of the gestures than an evaluation that distributes windows from each repetition between both partitions. Because only one subject and Exercise 1 are considered, the measured 74.26% accuracy should be interpreted as within-subject, repetition-disjoint performance, not as an estimate of cross-participant generalization. These conditions define the scope of the reported findings.

In summary, the complete pipeline—from twelve-channel EMG conditioning to 120-dimensional feature extraction and four-model evaluation—was executed on the recorded dataset. Random forest achieved 74.26% accuracy and 75.17% macro-F1 over seventeen movement classes, with class-wise performance and confusion patterns reported to characterize the recognition results.